

Notes on the proofs in *Sets Reconstructible with Medial Axis* prepared for a formalization in Lean 4 / mathlib*

June 2026

Abstract

We are formalizing Theorem 6 of [1] in the Lean 4 proof assistant, on top of the `mathlib` library. Machine checking forces every implicit step of a proof to be made fully explicit, and these notes record what that process surfaced. Concretely, we document: (i) one classical background fact which the paper uses through a citation (the Motzkin–Fremlin inclusion $C_X \subseteq \overline{M_X}$) and for which we give a self-contained elementary proof that avoids Brouwer’s fixed point theorem (which is not available in `mathlib`); (ii) a small toolbox of lemmas (a “ball membership along a ray” criterion and an “inflation dichotomy”) through which all the ball-inflation arguments of the paper factor; (iii) a reorganization in which Propositions 3 and 4 of [1] merge into a single statement whose proof does not use Motzkin’s theorem on Chebyshev sets at all, and turns out to be entirely free of limit arguments; (iv) a complete replacement for the proof of the forward direction of Theorem 6, where the argument sketched in [1] has a genuine gap. None of this affects the validity of the *statements* in [1]: Theorem 6 is correct as stated. The notes are written in ordinary mathematical language and require no knowledge of Lean. As of this version, every proof below *except* Appendix A has been machine-verified: Theorem 6 is fully checked in Lean modulo the single classical input (1).

1 Setting, notation, and the result being formalized

Throughout, E is a finite-dimensional real inner product space (the formalization is carried out at this level of generality; $E = \mathbb{R}^n$ is the motivating case), and $X \subseteq E$ is a nonempty closed set. Following [1] we write

$$d(a) = d(a, X) = \inf_{x \in X} \|a - x\|, \quad m(a) = \{x \in X \mid \|a - x\| = d(a)\},$$

$\mathbb{B}(a, r)$ for the *open* ball, and

$$M_X = \{a \in E \mid \#m(a) > 1\}, \quad C_X = \{\text{centres of inclusion-maximal open balls contained in } E \setminus X\}.$$

A point $p \in E \setminus X$ is *reconstructible* if $p \in \bigcup_{a \in M_X} \mathbb{B}(a, d(a))$, and $E \setminus X$ is reconstructible if all its points are. A hyperplane L is *supporting* for a set S if $S \cap L \neq \emptyset$ and S lies in one of

*Notes drafted with the assistance of Claude (Anthropic) during the planning stage of the formalization. All arguments below were re-derived and checked by hand at the level of detail a proof assistant requires; remaining errors are ours.

the two closed half-spaces bounded by L ; then L^+ denotes the open half-space disjoint from S . We represent hyperplanes by a unit normal: $L = \{x \mid \langle \eta, x \rangle = c\}$ with $S \subseteq \{\langle \eta, \cdot \rangle \leq c\}$ and $L^+ = \{\langle \eta, \cdot \rangle > c\}$. We call a supporting hyperplane L of $\overline{\text{conv}} X$ *defective* if $X \cap L \neq \overline{\text{conv}} X \cap L$; equivalently (since $X \cap L \subseteq \overline{\text{conv}} X \cap L$ always) if there exists a *defect witness*

$$w \in (\overline{\text{conv}} X \cap L) \setminus X.$$

The target is:

Theorem 1.1 ([1, Theorem 6]). *Let $X \subseteq E$ be closed and nonempty, and let $\mathcal{L}$ be the family of all defective supporting hyperplanes of $\overline{\text{conv}} X$. Then $E \setminus X$ is reconstructible if and only if*

$$E \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_{L \in \mathcal{L}} L^+.$$

(The paper says “let $\mathcal{L}$ be a family. . .”; for the forward implication $\mathcal{L}$ must be the family of *all* defective supporting hyperplanes, and we formalize the statement in that form.)

Summary of findings

Result in [1]	Status in the formalization
Lemma 1	Proved; the step “ c can be approximated by points of M_X ” is the Motzkin–Fremlin theorem $C_X \subseteq \overline{M_X}$, which we must prove ourselves (§3, Appendix A); in the first stage of the formalization this single classical fact is assumed.
Proposition 2	Proved; one definition repaired (Remark 4.1) and Carathéodory’s theorem is not needed (Remark 4.3).
Propositions 3, 4	Merged into one statement (Theorem 5.1) with a shorter proof: Motzkin’s theorem on Chebyshev sets, the cone at infinity, and the two-nearest-points analysis are no longer used; moreover the proof is entirely limit-free — a single explicit value of the lift parameter suffices (Remark 5.2).
Corollary 5	As written it invokes the <i>proof</i> (not the statement) of Proposition 4 and does not literally follow from it; we give a direct “one-shot” proof needing no limits (§6).
Theorem 6, “ $\Leftarrow$ ”	Assembly of the above.
Theorem 6, “ $\Rightarrow$ ”	The sketch in [1] has a gap (the segment $]p, q[$ may miss $\overline{\text{conv}} X$ entirely, and a separating hyperplane through the entry point need not exist); we give a complete replacement proof via the metric projection onto $\overline{\text{conv}} X$ (§7.2).

2 Toolbox

All inflation arguments in [1] grow a one-parameter family of balls that share a tangency point. The following trivial identity is the workhorse that replaces every limit computation “ p eventually falls into the growing balls” by an explicit inequality. (In a formalization such a reduction matters enormously; but we believe it also makes the paper proofs shorter.)

Lemma 2.1 (ball membership along a ray). *Let $x, p \in E$, let v be a unit vector and $\rho > 0$. Then*

$$p \in \mathbb{B}(x + \rho v, \rho) \iff 2\rho \langle v, p - x \rangle > \|p - x\|^2.$$

Proof. Expand $\|p - x - \rho v\|^2 < \rho^2$. □

Lemma 2.2 (tangent families). *Fix $x \in E$ and a unit vector v , and write $\mathbb{B}_t := \mathbb{B}(x + tv, t)$ for $t > 0$ (the family of open balls tangent to the hyperplane through x with normal v , at the point x). Then*

- (a) $0 < s \leq t \implies \mathbb{B}_s \subseteq \mathbb{B}_t$;
- (b) $\bigcup_{0 < t < T} \mathbb{B}_t = \mathbb{B}_T$ for every $T > 0$;
- (c) $\bigcup_{t > 0} \mathbb{B}_t = \{y \mid \langle v, y - x \rangle > 0\}$, the open half-space.

Proof. All three are immediate from Lemma 2.1: membership of y in $\mathbb{B}_t$ reads $2t \langle v, y - x \rangle > \|y - x\|^2$, in which the left side is nonnegative-increasing in t once $\langle v, y - x \rangle > 0$, and which for $y \neq x$ can hold only if $\langle v, y - x \rangle > 0$. □

We record the elementary facts about maximal balls; recall a ball $\mathbb{B}(c, r) \subseteq E \setminus X$, $r > 0$, is *maximal* if $\mathbb{B}(c, r) \subseteq \mathbb{B}(c', r') \subseteq E \setminus X$ (with $r' > 0$) forces $c' = c$, $r' = r$.

Lemma 2.3. *Let X be closed and nonempty.*

- (a) *For balls in a normed space, $\mathbb{B}(c, r) \subseteq \mathbb{B}(c', r')$ iff $\|c - c'\| \leq r' - r$.*
- (b) *A maximal ball centred at c has radius $d(c)$; thus $C_X = \{c \mid \mathbb{B}(c, d(c)) \text{ is maximal}\}$.*
- (c) $M_X \subseteq C_X$.
- (d) *If $a \in X$ then $m(a) = \{a\}$; hence $M_X \cap X = \emptyset$ and $d(a) > 0$ for $a \in M_X \cup C_X$.*

Proof sketch. (b) $\mathbb{B}(c, r) \subseteq E \setminus X$ iff $r \leq d(c)$. (c) Let $a \in M_X$ with distinct $x, y \in m(a)$ and suppose $\mathbb{B}(a, d(a)) \subseteq \mathbb{B}(c', r') \subseteq E \setminus X$. Since $x \in X$, $x \notin \mathbb{B}(c', r')$, so $\|x - c'\| \geq r'$; on the other hand by (a), $\|x - c'\| \leq \|x - a\| + \|a - c'\| \leq d(a) + (r' - d(a)) = r'$. So equality holds in the triangle inequality, which in a Euclidean (strictly convex) space forces c' to lie on the ray from x through a , at distance $r' - d(a)$ beyond a ; the same holds with y in place of x . If $r' > d(a)$ the two descriptions of c' give $(a - x)/d(a) = (a - y)/d(a)$, i.e. $x = y$, a contradiction; hence $r' = d(a)$ and $c' = a$. □

The next lemma is the common engine behind Propositions 2–4 and Corollary 5 of [1]. It is the precise form of “inflate a ball keeping the tangency point until it either becomes maximal or sweeps out a half-space”.

Lemma 2.4 (inflation dichotomy). *Let X be closed and nonempty, $x_0 \in X$, v a unit vector, and suppose $\mathbb{B}(x_0 + t_0 v, t_0) \cap X = \emptyset$ for some $t_0 > 0$. Put*

$$T := \sup\{t > 0 \mid \mathbb{B}(x_0 + tv, t) \cap X = \emptyset\} \in [t_0, \infty].$$

- (i) *If $T = \infty$, then $X \cap \{y \mid \langle v, y - x_0 \rangle > 0\} = \emptyset$.*
- (ii) *If $T < \infty$, then $c := x_0 + Tv \in C_X$, with $d(c) = T$ and $x_0 \in m(c)$.*
- (iii) *If moreover $p \notin X$, $x_0 \in m(p)$ and $v = (p - x_0)/d(p)$, then $t_0 = d(p)$ qualifies, and in case (ii) one has $p \in \mathbb{B}(c, d(c))$.*

Proof. By Lemma 2.2(a) the set of admissible t is an interval, so every $t < T$ is admissible. (i) is Lemma 2.2(c). For (ii): by Lemma 2.2(b), $\mathbb{B}(c, T) = \bigcup_{t < T} \mathbb{B}_t$ is disjoint from X , so $d(c) \geq T$; since $x_0 \in X$ and $\|c - x_0\| = T$ we get $d(c) = T$ and $x_0 \in m(c)$. Maximality: if $\mathbb{B}(c, T) \subseteq \mathbb{B}(c', r') \subseteq E \setminus X$ then, exactly as in the proof of Lemma 2.3(c), $\|x_0 - c'\| = r'$ with equality in the triangle inequality through c , so $c' = x_0 + r'v$; if $r' > T$ this contradicts the definition of T , hence $r' = T$, $c' = c$. For (iii): $\mathbb{B}(x_0 + d(p)v, d(p)) = \mathbb{B}(p, d(p))$ is disjoint from X , so $T \geq d(p)$; and by Lemma 2.1, $p \in \mathbb{B}(c, T)$ iff $2T \langle v, p - x_0 \rangle = 2T d(p) > d(p)^2$, which holds because $T \geq d(p) > \frac{1}{2}d(p)$. $\square$

3 Lemma 1 and the Motzkin–Fremlin theorem

The proof of [1, Lemma 1] uses that every $c \in C_X$ “can be approximated by a sequence of points $x_n \in M_X$ ”, i.e.

$$M_X \subseteq C_X \subseteq \overline{M_X}, \quad (1)$$

which the paper quotes from the literature (Motzkin; Fremlin [2]). For the formalization this inclusion must itself be proved, and it sits on the critical path of *everything*: all the reconstruction arguments produce centres of maximal balls, and a maximal ball may touch X at a single point, so the passage from C_X back to M_X is genuinely needed and genuinely nontrivial. The textbook proofs use Brouwer’s fixed point theorem or degree theory, neither of which is currently available in `mathlib`. In Appendix A we give a self-contained proof of $C_X \subseteq \overline{M_X}$ which uses nothing beyond compactness and an explicit Euler-polygon construction; we believe it may be of independent interest. (In the first stage of the formalization, (1) is assumed as a clearly marked placeholder and everything else is proved unconditionally; in the second stage we formalize Appendix A.)

Given (1), Lemma 1 of [1] is straightforward, exactly as in the paper:

Lemma 3.1 ([1, Lemma 1]). *Let $p \in E \setminus X$. Then p is reconstructible if and only if there is $c \in C_X$ with $p \in \mathbb{B}(c, d(c))$.*

Proof. $(\Rightarrow) M_X \subseteq C_X$ by Lemma 2.3(c). $(\Leftarrow)$ By (1) pick $a_n \in M_X$, $a_n \rightarrow c$. Since d is 1-Lipschitz, $d(a_n) \rightarrow d(c)$ and $\|p - a_n\| \rightarrow \|p - c\| < d(c)$, so $\|p - a_n\| < d(a_n)$ for large n . $\square$

4 Proposition 2

Remark 4.1 (a repaired definition). The proof of [1, Proposition 2] introduces, for $p_t := t(p - x_p) + x_p$,

$$r := \sup\{t > 0 \mid \mathbb{B}(p_t, d(p_t)) \cap X = \emptyset\},$$

and argues by cases on $r < \infty$. As literally written the defining condition is vacuous: $\mathbb{B}(a, d(a)) \cap X = \emptyset$ holds for *every* a , so $r = \infty$ always. What is meant (and what the surrounding text uses, e.g. “it is the union of the family of balls”) is the family of balls *tangent to X at x_p* , i.e. $\mathbb{B}(p_t, t d(p))$, equivalently the condition $d(p_t) = t d(p)$. Lemma 2.4 is exactly this corrected statement, packaged once and for all; the same correction applies wherever the “directional reaching radius” $r_v(x) := \sup\{\rho \geq 0 \mid \mathbb{B}(\rho v + x, \rho) \cap X = \emptyset\}$ of [1] is used, which matches our T in Lemma 2.4.

Proposition 4.2 ([1, Proposition 2]). *If $p \in \text{conv } X \setminus X$, then p is reconstructible.*

Proof. Pick $x_p \in m(p)$ (nonempty: X closed nonempty, E finite-dimensional) and let $v = (p - x_p)/d(p)$. Apply Lemma 2.4(iii). If $T < \infty$ we obtain $c \in C_X$ with $p \in \mathbb{B}(c, d(c))$, and Lemma 3.1 concludes. If $T = \infty$, then $X \subseteq K := \{y \mid \langle v, y - x_p \rangle \leq 0\}$. But K is convex, so $\text{conv } X \subseteq K$, and $p \in \text{conv } X$ gives $\langle v, p - x_p \rangle \leq 0$; since $\langle v, p - x_p \rangle = d(p) > 0$, this is a contradiction. $\square$

Remark 4.3. The paper's proof invokes Carathéodory's lemma to write p as a convex combination of points of X and then computes. This is not needed: the half-space K is convex and contains X , so it contains $\text{conv } X \ni p$ — the computation with the convex combination in [1] is precisely the proof of this containment in the special case. (In the formalization this saves a dependency on the Carathéodory machinery.)

5 A unified Proposition 3/4

This section contains the main reorganization. Recall [1, Proposition 3]: if L is a supporting hyperplane of $\text{conv } X$ with $X \cap L$ *non-convex*, then every $p \in L^+$ is reconstructible; and [1, Proposition 4]: the same conclusion when L is a supporting hyperplane of $\overline{\text{conv}} X$ with $X \cap L \neq \overline{\text{conv}} X \cap L$. The proof of Proposition 3 in [1] starts from Motzkin's theorem (a closed non-convex set has a nonempty medial axis, applied to $X \cap L$ inside L) to obtain a point q with two nearest points $a \neq b$ in $X \cap L$, and the proof of Proposition 4 reduces its convex case to a separate ladder construction.

Two observations let us merge the two propositions and shorten the proof considerably.

- (A) The hypothesis of Proposition 3 implies that of Proposition 4: if $X \cap L$ is non-convex, then $X \cap L \subsetneq \text{conv}(X \cap L) \subseteq \overline{\text{conv}} X \cap L$, so L is defective. Hence a single statement with the “defective” hypothesis covers both.
- (B) In the ladder construction, the role played in [1] by the pair a, b of nearest points (Proposition 3) or by approximating convex combinations a_ν, b_ν (Proposition 4) can be played directly by a *defect witness* $w \in (\overline{\text{conv}} X \cap L) \setminus X$: in the “infinite radius” case the half-space produced by Lemma 2.4(i) is closed and convex, hence contains $\overline{\text{conv}} X \ni w$, and evaluating the half-space inequality at w gives an immediate contradiction. *Consequently Motzkin's theorem is not needed anywhere in the proof of Theorem 6.* (It remains an interesting target in its own right; see Remark 5.3.)

Theorem 5.1 (unified Propositions 3 and 4). *Let X be closed and nonempty and let $L = \{x \mid \langle \eta, x \rangle = c\}$, $\|\eta\| = 1$, satisfy*

$$X \subseteq \{x \mid \langle \eta, x \rangle \leq c\}, \quad \exists w \in \overline{\text{conv}} X \cap L, \quad w \notin X.$$

Then every p with $\langle \eta, p \rangle > c$ is reconstructible.

Proof. Translating, we may assume $c = 0$; write $\gamma := \langle \eta, p \rangle > 0$, and fix *once and for all* the lift parameter

$$t := \frac{\|p - w\|^2}{2\gamma} + 1, \quad \text{so that} \quad \|p - w\|^2 = 2t\gamma - 2\gamma.$$

Consider the lifted point $w_t := w + t\eta$; since $\langle \eta, w_t \rangle = t > 0$ while $X \subseteq \{\langle \eta, \cdot \rangle \leq 0\}$, we have $w_t \notin X$, so

$$x_t \in m(w_t), \quad D := d(w_t) = \|w_t - x_t\| > 0, \quad v := \frac{w_t - x_t}{D}$$

are well defined. Abbreviate

$$S := w - x_t, \quad P := p - w, \quad \beta := -\langle \eta, x_t \rangle \geq 0$$

(the inequality because $x_t \in X$). Everything rests on two instances of the same expansion, using $w_t - x_t = t\eta + S$, $\langle \eta, S \rangle = \beta$ and $\langle \eta, P \rangle = \gamma$:

$$\langle w_t - x_t, w - x_t \rangle = \langle t\eta + S, S \rangle = t\beta + \|S\|^2, \quad (2)$$

$$N := \langle w_t - x_t, p - x_t \rangle = \langle t\eta + S, P + S \rangle = t\gamma + t\beta + \langle S, P \rangle + \|S\|^2. \quad (3)$$

The ball $\mathbb{B}(w_t, D)$ is the tangent ball of radius D at x_t in the direction v and avoids X ; apply Lemma 2.4.

The case $T = \infty$ is impossible. By Lemma 2.4(i), $X \subseteq K := \{y \mid \langle v, y - x_t \rangle \leq 0\}$; K is closed and convex, so $\overline{\text{conv}} X \subseteq K$, and in particular $\langle w_t - x_t, w - x_t \rangle \leq 0$. By (2) this forces $t\beta + \|S\|^2 \leq 0$, hence $S = 0$, i.e. $x_t = w$ — contradicting $x_t \in X$, $w \notin X$.

The case $T < \infty$, and the capture of p . By Lemma 2.4(ii), $c_T := x_t + Tv \in C_X$ with $d(c_T) = T \geq D$. By Lemmas 2.1 and 3.1 it remains to check

$$\|p - x_t\|^2 < 2T \langle v, p - x_t \rangle = \frac{2T}{D} N. \quad (4)$$

First, $N \geq 0$: by Cauchy–Schwarz $\langle S, P \rangle \geq -\|S\| \|P\|$, by $(\|S\| - \frac{1}{2} \|P\|)^2 \geq 0$ we have $\|S\|^2 - \|S\| \|P\| \geq -\frac{1}{4} \|P\|^2$, and $\|P\|^2 \leq 2t\gamma$ by the choice of t ; hence

$$N \geq t\gamma + t\beta - \frac{1}{4} \|P\|^2 \geq t\gamma - \frac{1}{2} t\gamma > 0.$$

Consequently $\frac{2T}{D} N \geq 2N$, and it suffices to compare $2N$ with $\|p - x_t\|^2 = \|P + S\|^2 = \|P\|^2 + 2\langle S, P \rangle + \|S\|^2$. The cross terms cancel *exactly* against (3):

$$2N - \|p - x_t\|^2 = 2t\gamma + 2t\beta + \|S\|^2 - \|P\|^2 \geq 2t\gamma - \|P\|^2 = 2\gamma > 0.$$

This proves (4); so $p \in \mathbb{B}(c_T, d(c_T))$ with $c_T \in C_X$, and Lemma 3.1 finishes the proof. $\square$

Remark 5.2 (on the final limit step in [1]). The proofs of Propositions 3 and 4 in [1] conclude with: “For any p in L^+ and a sequence of points $c_n \in C_X$ with $\|c_n\| \rightarrow \infty$, $c_n/\|c_n\| \rightarrow \eta$ we will have $d(c_n) > \|p - c_n\|$ for n large enough.” The two stated properties of the sequence are not by themselves sufficient for the conclusion: writing $c_n = s_n u_n$ with $s_n \rightarrow \infty$, $u_n \rightarrow \eta$, the available lower bound $d(c_n) \geq \langle \eta, c_n \rangle$ beats $\|p - c_n\| \approx s_n - \langle p, u_n \rangle$ only if $s_n(1 - \langle u_n, \eta \rangle)$ stays small, which the hypotheses do not guarantee for an *arbitrary* such sequence. The sequences actually *constructed* in the paper do satisfy the conclusion, but proving it requires retaining the tangency data of the maximal balls (tangency point and radius), which is what the proof above does via Lemma 2.1. This is a typical example of a step that is morally right but that a proof assistant (correctly) refuses without the extra data. Strikingly, once that data is kept *exactly*, all asymptotics evaporate: the cross terms $\langle S, P \rangle$ cancel between the two sides of (4), a single explicit value of t suffices, and neither the limit $t \rightarrow \infty$ nor the coordinate estimates for x_t in the proof of [1, Proposition 4] (nor even the nearest point of w itself) are needed. We found this simplification only while machine-checking the proof.

Remark 5.3 (status of Proposition 3 and Motzkin’s theorem). By observation (A) above, [1, Proposition 3] is a special case of Theorem 5.1; the formalization therefore does not state it separately, and Motzkin’s theorem (closed Chebyshev subsets of $\mathbb{R}^n$ are convex) leaves the dependency graph of Theorem 6 entirely. We note that once $C_X \subseteq \overline{M_X}$ is available (Appendix A), Motzkin’s theorem falls out for free, by the argument of Proposition 4.2: if $M_X = \emptyset$ then $C_X = \emptyset$, so for any $p \in \text{conv } X \setminus X$ the dichotomy (Lemma 2.4(iii)) cannot terminate at a maximal ball, hence produces a half-space through a nearest point separating p from X — contradicting $p \in \text{conv } X$ as in Proposition 4.2. Hence a non-convex closed set has points with at least two nearest points. We plan to include this as a corollary in the formalization since it is of independent interest.

6 Corollary 5: a one-shot proof

Remark 6.1. [1, Corollary 5] (every $p \in \overline{\text{conv}} X \setminus X$ is reconstructible) is justified by “any supporting hyperplane L of $\overline{\text{conv}} X$ with $p \in L$ will raise a sequence of balls as in the proof of Proposition 4”. Two points need attention. First, this invokes the *construction* inside the proof of Proposition 4 rather than its statement; moreover the balls of that construction capture the open half-space L^+ , while p lies *on* L , so the capture of p itself needs a separate argument. Second, the existence of a supporting hyperplane through a boundary point requires a (standard but nonempty) argument when $\overline{\text{conv}} X$ has empty interior. Both points are easy to settle, and in fact no limit $t \rightarrow \infty$ is needed at all:

Corollary 6.2 ([1, Corollary 5]). *Every $p \in \overline{\text{conv}} X \setminus X$ is reconstructible.*

Proof. If $p \in \text{conv } X$, apply Proposition 4.2. So assume $p \in \overline{\text{conv}} X \setminus \text{conv } X$.

A supporting hyperplane through p . If $\text{int conv } X \neq \emptyset$ then $\text{int } \overline{\text{conv}} X = \text{int conv } X \subseteq \text{conv } X$, so p lies on the boundary of the closed convex set $\overline{\text{conv}} X$ and a supporting hyperplane $L = \{\langle \eta, \cdot \rangle = c\}$ through p exists by standard separation. If $\text{int conv } X = \emptyset$, then $\overline{\text{conv}} X$ lies in a proper affine subspace A ; any unit η orthogonal to the direction of A gives a hyperplane through p containing $\overline{\text{conv}} X$, which is in particular supporting. In both cases, after translation $c = 0$,

$$X \subseteq \{\langle \eta, \cdot \rangle \leq 0\}, \quad \langle \eta, p \rangle = 0, \quad p \in \overline{\text{conv}} X, \quad p \notin X.$$

One-shot capture. Fix any $t > 0$ (say $t = 1$) and run the construction of Theorem 5.1 with defect witness $w := p$ and target point p itself: let $x_t \in m(p + t\eta)$, $D = d(p + t\eta)$, $v = (p + t\eta - x_t)/D$, and $S := p - x_t \neq 0$. The case $T = \infty$ is impossible exactly as before (it would force $x_t = w = p \notin X$). If $T < \infty$ we get $c_T \in C_X$, $d(c_T) = T \geq D$, and the membership criterion for p reads

$$2T \langle v, p - x_t \rangle \geq \frac{2T}{D} (t \langle \eta, S \rangle + \|S\|^2) \geq 2\|S\|^2 > \|S\|^2 = \|p - x_t\|^2,$$

using $\langle \eta, S \rangle = -\langle \eta, x_t \rangle \geq 0$ and $T \geq D$. So $p \in \mathbb{B}(c_T, d(c_T))$, and Lemma 3.1 concludes. (No limit in t was required; in the Lean development this case and Theorem 5.1 are two instances of one lemma, with the hypotheses $\langle \eta, p \rangle \geq c$ and $\|p - w\|^2 \leq 2t(\langle \eta, p \rangle - c)$.) $\square$

7 Theorem 6

7.1 The backward implication

Suppose $E \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_{L \in \mathcal{L}} L^+$. Every $p \in E \setminus X$ then lies either in $\overline{\text{conv}} X$ (Corollary 6.2) or in L^+ for some defective supporting hyperplane L (Theorem 5.1); in both cases p is

reconstructible. □

7.2 The forward implication: a gap and its repair

The argument sketched in [1] is: given a reconstructible $p \notin \text{conv } X$ with $p \in \mathbb{B}(a, d(a))$, $a \in M_X$, pick $q \in m(a)$, connect p and q inside the ball, let $[\bar{a}, \bar{b}] \subseteq]p, q[$ be the part of the segment inside $\overline{\text{conv}} X$, observe $\bar{a} \in (\overline{\text{conv}} X \cap L) \setminus X$ for any supporting $L \ni \bar{a}$, and finally “pick L to be a hyperplane that separates p and $\overline{\text{conv}} X$ ”.

Two distinct difficulties arise when one tries to make this precise.

Example 7.1 (the interval may not exist). Let $X = \{(0, 1), (0, -1)\} \subseteq \mathbb{R}^2$. Then M_X is the horizontal axis; take $a = (1, 0)$, $d(a) = \sqrt{2}$, and $p = (2, 0) \in \mathbb{B}(a, \sqrt{2})$, $p \notin \overline{\text{conv}} X = \{0\} \times [-1, 1]$. For either choice of $q \in m(a) = \{(0, \pm 1)\}$ the open segment $]p, q[$ meets the vertical axis only in the limit q itself, so $]p, q[\cap \overline{\text{conv}} X = \emptyset$: the interval $[\bar{a}, \bar{b}]$ of the proof is empty. (The theorem itself is fine here: $L = \{x_1 = 0\}$ is defective — its hull section is the segment, its X -section two points — and $p \in L^+$.)

Example 7.2 (grazing). Even when the entry point $\bar{a}$ exists, there need not be *any* supporting hyperplane through $\bar{a}$ having p strictly on its open side. If the boundary of $\overline{\text{conv}} X$ near $\bar{a}$ looks like the parabola $y = x^2$ near the origin and $p = (1, 0)$, the unique supporting line at $\bar{a} = (0, 0)$ is $\{y = 0\}$, which contains p rather than separating it. So “a hyperplane through $\bar{a}$ separating p and $\overline{\text{conv}} X$ ” may simply not exist, and the two requirements on L (defect witnessed at $\bar{a} \in L$; $p \in L^+$) cannot in general be met by one hyperplane chosen this way.

The repair below produces the defect witness and the strict separation *from the same metric-projection construction*, using the second nearest point of a (which the sketch in [1] never uses — note that Example 7.1 shows one tangency point is not enough!). Recall that in a finite-dimensional space the metric projection $\pi : E \rightarrow \overline{\text{conv}} X$ onto a closed convex set is well defined (existence and uniqueness of the nearest point), and is characterized by

$$z = \pi(b) \iff z \in \overline{\text{conv}} X \text{ and } \forall x \in \overline{\text{conv}} X : \langle b - z, x - z \rangle \leq 0. \quad (5)$$

Theorem 7.3 (forward implication of Theorem 1.1). *Let X be closed and nonempty and let $p \in E \setminus X$ be reconstructible with $p \notin \overline{\text{conv}} X$. Then there is a defective supporting hyperplane L of $\overline{\text{conv}} X$ with $p \in L^+$.*

Proof. Pick $a \in M_X$ with $p \in \mathbb{B} := \mathbb{B}(a, d(a))$ and two distinct points $q, q' \in m(a)$, and set

$$q'' := \tfrac{1}{2}(q + q').$$

By the parallelogram identity, $\|a - q''\|^2 = d(a)^2 - \frac{1}{4} \|q - q'\|^2 < d(a)^2$, so q'' lies in the *open* ball $\mathbb{B} \subseteq E \setminus X$; in particular $q'' \notin X$, while $q'' \in \text{conv } X \subseteq \overline{\text{conv}} X$.

Both endpoints of the segment $[p, q'']$ lie in the open convex set $\mathbb{B}$, hence so does the whole segment. Since $q'' \in \overline{\text{conv}} X$ and $p \notin \overline{\text{conv}} X$, and $\overline{\text{conv}} X$ is closed, the segment has a *first* point of $\overline{\text{conv}} X$ as seen from p : there is $\bar{a} \in [p, q''] \cap \overline{\text{conv}} X$ with

$$[p, \bar{a}] \cap \overline{\text{conv}} X = \emptyset, \quad \bar{a} \neq p.$$

Note $\bar{a} \in \mathbb{B}$, so $\bar{a} \notin X$. Let $u := (p - \bar{a}) / \|p - \bar{a}\|$.

For $b \in [p, \bar{a}]$ write $z_b := \pi(b)$ and $g_b := \|b - z_b\| = \text{dist}(b, \overline{\text{conv}} X) > 0$. Since $\bar{a} \in \overline{\text{conv}} X$,

$$\|z_b - \bar{a}\| \leq \|z_b - b\| + \|b - \bar{a}\| \leq 2 \|b - \bar{a}\| \xrightarrow{b \rightarrow \bar{a}} 0.$$

As $\bar{a} \notin X$ and X is closed, we may fix $b \in [p, \bar{a}[$ with $z_b \notin X$.

Now define the hyperplane by the unit normal $v := (b - z_b)/g_b$ and level $c := \langle v, z_b \rangle$, i.e. $L := \{x \mid \langle v, x \rangle = c\}$. We check the three required properties.

L supports $\overline{\text{conv}} X$. By (5), $\langle v, x - z_b \rangle \leq 0$ for all $x \in \overline{\text{conv}} X$, i.e. $\overline{\text{conv}} X \subseteq \{\langle v, \cdot \rangle \leq c\}$; and $z_b \in \overline{\text{conv}} X \cap L$. (A fortiori $X \subseteq \{\langle v, \cdot \rangle \leq c\}$.)

L is defective. $z_b \in (\overline{\text{conv}} X \cap L) \setminus X$ by the choice of b .

$p \in L^+$. Since b lies on the segment from $\bar{a}$ to p we have $b - \bar{a} = \|b - \bar{a}\| u$ and $p - b = \|p - b\| u$. Using $\langle v, \bar{a} \rangle \leq c$ (as $\bar{a} \in \overline{\text{conv}} X$),

$$\langle v, u \rangle = \frac{\langle v, b - \bar{a} \rangle}{\|b - \bar{a}\|} = \frac{\langle v, b - z_b \rangle + \langle v, z_b - \bar{a} \rangle}{\|b - \bar{a}\|} \geq \frac{g_b + 0}{\|b - \bar{a}\|} > 0,$$

hence

$$\langle v, p \rangle - c = \langle v, p - z_b \rangle = \langle v, p - b \rangle + \langle v, b - z_b \rangle = \|p - b\| \langle v, u \rangle + g_b > 0. \quad \square$$

Remark 7.4. The proof uses only the metric projection onto a closed convex set and no separation theorems; it is insensitive to whether $\overline{\text{conv}} X$ has empty interior. In Example 7.1 it produces $q'' = (0, 0)$, $\bar{a} = (0, 0)$, $z_b = (0, 0)$ and the defective vertical line, as expected. Combining both implications with Corollary 6.2 and Theorem 5.1 yields Theorem 1.1.

8 Remarks on the formalization

The development has been carried out in Lean 4 over `mathlib`, for an arbitrary finite-dimensional real inner product space; the definitions (m , M_X , C_X , reconstructibility) are made for general metric spaces, with the geometric results at the inner-product level. The structure mirrors these notes: the toolbox of §2, then Lemma 3.1, Proposition 4.2, Theorem 5.1, Corollary 6.2, and the two implications of Theorem 1.1. As of this version, *all of the above is machine-checked* (≈ 1100 lines in three files; the main statement is `Metric.reconstructible_iff`). The single classical input (1) ($C_X \subseteq \overline{M_X}$) is assumed as an explicitly marked placeholder, and Lean’s axiom tracker confirms it is the *only* unproved dependency of Theorem 1.1; formalizing Appendix A to discharge it is the next step. The forward implication of Theorem 1.1 (§7.2) is checked with no assumptions at all. We will gladly share the Lean sources; comments on the mathematics in these notes, and in particular on Appendix A, are very welcome.

A A Brouwer-free proof that $C_X \subseteq \overline{M_X}$

Throughout, $X \subseteq E$ is closed and nonempty, E finite-dimensional. We prove: *the centre of any maximal ball lies in the closure of M_X* . The strategy: if $c \in C_X$ had a neighbourhood free of M_X , the nearest-point map would be single-valued, hence continuous, near c ; we then inflate the maximal ball by following an explicit Euler polygon that always moves directly away from the current nearest point, gaining distance from X at unit rate up to an error controlled by the (uniform) continuity of the nearest-point map; a compactness argument upgrades the almost-unit rate to unit rate, producing a ball that strictly contains the maximal ball — a contradiction.

Lemma A.1. *Let $K \subseteq E \setminus X$ be compact with $K \cap M_X = \emptyset$. Then every $a \in K$ has a unique nearest point $P(a)$, and P is uniformly continuous on K .*

Proof. $m(a) \neq \emptyset$ by closedness of X and finite dimension; single-valuedness on K is the assumption. The graph of m is closed (if $a_n \rightarrow a$, $x_n \in m(a_n)$, $x_n \rightarrow x$, then $x \in X$ and $\|a - x\| = \lim d(a_n) = d(a)$ by continuity of d), and m is locally bounded ($\|x\| \leq \|a\| + d(a)$ for $x \in m(a)$). A single-valued, locally bounded, closed-graph map into a finite-dimensional space is continuous (any subsequential limit of $P(a_n)$ lies in $m(a) = \{P(a)\}$), and continuity on a compact set is uniform. $\square$

Lemma A.2 (quadratic gain estimate). *Let $a \notin X$, $x_a \in m(a)$, $v := (a - x_a)/d(a)$, $h > 0$, and let $y \in m(a + hv)$ be arbitrary. Then*

$$d(a + hv)^2 \geq (d(a) + h)^2 - \frac{h \|x_a - y\|^2}{d(a)}.$$

Proof. Since y realizes the distance, $d(a + hv)^2 = \|a + hv - y\|^2 = \|a - y\|^2 + 2h \langle v, a - y \rangle + h^2$. Now $\|a - y\| \geq d(a)$, and by the polarization identity

$$\langle a - x_a, a - y \rangle = \frac{1}{2} (\|a - x_a\|^2 + \|a - y\|^2 - \|x_a - y\|^2) \geq d(a)^2 - \frac{1}{2} \|x_a - y\|^2,$$

so $\langle v, a - y \rangle \geq d(a) - \|x_a - y\|^2 / (2d(a))$. Substituting gives the claim. $\square$

Theorem A.3. $C_X \subseteq \overline{M_X}$.

Proof. Let $c \in C_X$, so $R := d(c) > 0$ and $\mathbb{B}(c, R)$ is maximal. Suppose, for contradiction, that $\overline{\mathbb{B}}(c, \varepsilon) \cap M_X = \emptyset$ for some $\varepsilon \in (0, R)$. Write $\ell := \varepsilon/2$ and $m_0 := R - \ell > 0$. By Lemma A.1 the nearest-point map P is single-valued and uniformly continuous on $\overline{\mathbb{B}}(c, \varepsilon)$; let ω be a modulus, i.e. $\|P(a) - P(a')\| \leq \omega(h)$ whenever $a, a' \in \overline{\mathbb{B}}(c, \varepsilon)$, $\|a - a'\| \leq h$, with $\omega(h) \rightarrow 0$ as $h \rightarrow 0$.

The polygon. Fix $N \in \mathbb{N}$, $h := \ell/N$, and define $a_0 := c$ and

$$a_{k+1} := a_k + h v_k, \quad v_k := \frac{a_k - P(a_k)}{d(a_k)} \quad (k = 0, \dots, N-1).$$

By induction $\|a_k - c\| \leq kh \leq \ell$, so all nodes stay in $\overline{\mathbb{B}}(c, \ell)$ and $d(a_k) \geq R - \ell = m_0$. By Lemma A.2 applied at a_k (with $y = P(a_{k+1})$, $\|P(a_k) - P(a_{k+1})\| \leq \omega(h)$),

$$d(a_{k+1})^2 \geq (d(a_k) + h)^2 - \frac{h \omega(h)^2}{m_0}.$$

Using $\sqrt{A^2 - B} \geq A - B/A$ (valid for $0 \leq B \leq A^2$) with $A = d(a_k) + h \geq m_0$:

$$d(a_{k+1}) \geq d(a_k) + h - \frac{h \omega(h)^2}{m_0 (d(a_k) + h)} \geq d(a_k) + h \left(1 - \frac{\omega(h)^2}{m_0^2}\right),$$

provided h is small enough that $\omega(h)^2 \leq m_0^2$. Summing over the N steps,

$$d(a_N) \geq R + \ell (1 - \delta(h)), \quad \delta(h) := \frac{\omega(h)^2}{m_0^2} \xrightarrow{h \rightarrow 0} 0.$$

The limit ball. The endpoints $a_N = a_N^{(h)}$ lie in the compact set $\overline{\mathbb{B}}(c, \ell)$; choosing $h_j \rightarrow 0$ and passing to a convergent subsequence we obtain $w \in \overline{\mathbb{B}}(c, \ell)$ with (by continuity of d)

$$\|w - c\| \leq \ell, \quad d(w) \geq R + \ell.$$

Then $\mathbb{B}(c, R) \subseteq \mathbb{B}(w, d(w))$: for $z \in \mathbb{B}(c, R)$, $\|z - w\| \leq \|z - c\| + \ell < R + \ell \leq d(w)$. Since $\mathbb{B}(w, d(w)) \subseteq E \setminus X$, maximality of $\mathbb{B}(c, R)$ forces $(w, d(w)) = (c, R)$, contradicting $d(w) \geq R + \ell > R$. $\square$

Remark A.4. The proof uses neither Brouwer’s fixed point theorem nor existence theory for ODEs (Peano), the two standard tools here: the flow “always move away from your nearest point” is replaced by its Euler polygon, whose per-step loss is controlled by Lemma A.2, and the inexact gain rate $1 - \delta(h)$ is repaired in the limit by compactness. Together with $M_X \subseteq C_X$ (Lemma 2.3(c)) this gives (1), and, as noted in Remark 5.3, Motzkin’s theorem on Chebyshev sets follows as a corollary.

References

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- [3] The mathlib Community, *The Lean mathematical library*, <https://github.com/leanprover-community/mathlib4>.